

# RIDDHI GOSWAMI

+971-502563195 | [ridhgoswami@gmail.com](mailto:ridhgoswami@gmail.com) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## PROFILE

Computer Science student (CGPA: 9.98) awarded the UAE Golden Visa for academic excellence. Experienced in ML, AI, IoT, and software engineering. Proven leadership and technical delivery at Emirates Group, Jacobs, and ongoing internship at Esri. Skilled in software development and technical communication. Quick to grasp new technologies and concepts, with a strong ability to adapt and contribute in dynamic environments.

## EDUCATION

### **BITS Pilani - Dubai Campus**

*B.E. (Hons) Computer Science: CGPA 9.98 - First Rank (2022-2024)*

Dubai, UAE

*Expected Aug 2025*

### **The Indian High School**

*Grade 12 - Science Stream: Grade 98.6% - School Topper 2nd Position*

Dubai, UAE

*2021*

### **The Indian High School**

*Grade 10: Grade 97%*

Dubai, UAE

*2019*

## EXPERIENCE

### **Software Developer Intern**

*Esri*

Jan. 2025 – Present

*Sharjah, UAE*

- Contributed to the development and debugging of advanced geospatial tools in the Oriented Imagery (OI) division of ArcGIS Pro, working primarily in C++ with integration across .NET, WPF, and COM-based systems.
- Resolved critical bugs affecting UI bindings, image overlays, viewer resets, and Geoprocessing (GP) tool crashes, with deep investigation into event dispatching, map-view state, and metadata parsing.
- Enhanced GP tool workflow to support video and mosaic ingestion into the OI system by extracting geospatial metadata and transforming into compliant OI dataset.
- Worked on a tool to export depth imagery layers as georeferenced TIFFs, aligning outputs with image metadata and camera footprints for use in 3D analysis and reconstruction.
- Proposed future-facing testing strategies (CUIT and graphic harnesses) to enhance automated UI validation, and developed unit tests to strengthen code reliability.

### **Software Developer - Structure Team**

*Jacobs*

Jun. 2024 – Aug. 2024

*Dubai, UAE*

- Engineered dynamic desktop applications using WPF with .NET 8.0 and integrated PyShiny for automating complex structural design workflows.
- Applied advanced NLP techniques—including Seq2Seq models, BERT, and fuzzy string matching—to standardise and enforce enterprise-wide naming conventions.
- Developed proficiency in Agile methodologies and Azure DevOps, leveraging design patterns and clean code principles for scalable and maintainable solutions.

### **Revenue Optimization Intern**

*Emirates Group HQ*

Jun. 2023 – Aug. 2023

*Dubai, UAE*

- Developed machine learning models to accurately predict flight sell-out dates (MAE < 2 days) using high-volume datasets exceeding 500 million records.
- Engineered Big Data optimisations that significantly reduced ML model training time and improved pipeline efficiency.
- Designed interactive, data-rich dashboards with Plotly, Dash, and Streamlit to support real-time decision-making.
- Leveraged Snowflake for scalable data storage and fast analytical querying across distributed datasets.

## MISCELLANEOUS EXPERIENCE

**Management Lead**, Google Developer Groups - BPDC  
**Professional Assistant - Computer Architecture**, CS Dept. - BPDC  
**Event Manager**, IEEE BPDC  
**Software Team Member**, IFOR Drone and Robotics Research

*Sept. 2024 – Present*  
*Sept. 2024 – Jan. 2025*  
*Sept. 2023 – Aug. 2024*  
*Sept. 2023 – Aug. 2024*

## TECHNICAL SKILLS

**Languages:** C/C++, C#, Python, Java, HTML/CSS  
**Frameworks:** .NET 8.0, ReactJs, PyShiny, Dash, Streamlit  
**Data Processing:** Numpy, Pandas, Polars, MySQL, SnowflakeSQL, Excel  
**AI/ML:** Sklearn, OpenCV, TensorFlow, Keras, Spacy, NLTK, LLMs  
**Misc. Tools:** Git, Azure DevOps, Power BI, Arduino, ROS/ROS2

## SELECTED PROJECTS

### Dynamic Backdoor Attackers in Federated Learning

Jan. 2025 – Present

- \* Conducted empirical research on the resilience of Federated Learning models against diverse backdoor attacks using CIFAR-10 and SimpleCNN in non-IID settings.
- \* Identified the impact of trigger pattern consistency on attack stealth and effectiveness, contributing to insights on secure FL architecture design.
- \* Currently authoring a journal paper based on this research for academic publication.

### Automated Detection and Structural Health Monitoring

Nov. 2024 – Present

- \* Designed a Deep Learning-based pipeline for automated structural health monitoring of civil infrastructure.
- \* Trained custom ResNet, InceptionV3, and Transformer-based CoCa and RevCol models on a proprietary defect dataset for multi-class classification.
- \* Currently authoring a journal paper based on this research for academic publication.

### Voice-Controlled Mobile Bot

Jan. 2024 – Jan. 2025

- \* NLP + ROS workflow using WhisperAPI, GPT-4o, ElevenLabs, Bark.
- \* Real-time voice command recognition and robot mobility control.
- \* Presented and published my work in IEEE CINS 2024 conference: [DOI: 10.1109/CINS.2024.10864443](https://doi.org/10.1109/CINS.2024.10864443)

### AlphaDock: DNA-Protein Complex Predictor

Jan. 2024 – Jan. 2025

- \* Developed a computational biology framework for predicting DNA-protein docking configurations using TF binding affinity datasets.
- \* Analysed transcription factor interactions using UniProbe and CIS-BP databases for complex structure predictions.
- \* Presented as a final-year research project under faculty review for evaluation and distinction.

## CERTIFICATIONS AND ACCOLADES

Google x Kaggle Gen AI Intensive  
HarvardX CS50 AI  
NNs and Deep Learning - DeepLearning.AI  
ACM Datathon 1st Place  
ADU Datathon 2nd Place - WiDS  
ML Project Structuring - DeepLearning.AI  
Microsoft Learn AI Challenge  
HarvardX CS50x Cert.  
GDSC Ideathon Winner  
Postman API Student Leader

*Apr. 2025*  
*Jul. 2024*  
*Jul. 2024*  
*Mar. 2024*  
*Mar. 2024*  
*Jan. 2024*  
*Aug. 2023*  
*Jan. 2023*  
*Feb. 2023*  
*2022–Present*